Fourier-Based Spectral–Spatial Generator for Cross-Scene Hyperspectral Image Classification

Boshan Shi, Guo Cao, *Member, IEEE*, Youqiang Zhang, *Member, IEEE*, Yanbo Liu, and Kai Yang

***Abstract*— Domain generalization (DG) has shown significant potential for cross-scene hyperspectral image (HSI) classification, wherein a model is trained exclusively on the source domain (SD) and can be directly transferred to an unseen target domain (TD). Current DG-based methods focus only on expanding the distribution of SDs by randomizing the style of the entire HSI cube. They fail to account for the domain shift problem caused by the variance of spatial land-cover distribution (context semantics), which results in SD-specific patterns being overly emphasized during training and, consequently, limiting the generalizability. Moreover, such randomization on cubes may introduce undesirable artifacts, such as blurring or distortion, leading to semantically compromised samples. In this article, a Fourier-based spectral–spatial generator (FSSG) is proposed to generate diversified and robust generative domain (GD). Specifically, a Fourier disentanglement is developed to construct spectral expansion (SpeE) and spatial expansion (SpaE) from pixelwise and regionwise levels, respectively. In SpeE, the style information is transmitted across pixels in a privacy-protecting way, i.e., SD shares the semantic information with the GD. In SpaE, an effective continuous frequency space interpolation is employed to transmit the styles and semantics information across cubes, which enables GD to bridge interdomain gaps in both styles and context semantics. To further alleviate the overemphasis on SD-specific patterns, a relaxation procedure is integrated within an adversarial training based on a coarse-to-fine paradigm, which facilitates the HSI cubes to gain more robust context semantics. Extensive experiments and analyses, conducted with two baseline methods across three public datasets, demonstrate the superiority of the proposed approach. The codes will be available from the website:https://github.com/BoshanCj/FSSG.**

***Index Terms*— Cross-scene classification, domain adversarial learning, domain generalization (DG), frequency interpolation, hyperspectral images (HSIs).**

I. Introduction

HYPERSPECTRAL image (HSI) classification [1] aims to infer the label of each pixel in an HSI cube by utilizing spectral–spatial attributes. This is a fundamental problem in the field of computer vision. Over the past decade, HSI classification has made significant strides due to the availability of public datasets and the rapid evolution of deep learning [2], [3], providing strong support for understanding and addressing various issues [4], [5], [6], [7] in modern society.

Deep learning-based HSI classification methods are primarily divided into three categories: spectral feature networks, spatial feature networks, and spectral–spatial feature networks. *Spectral feature networks* focus solely on spectral information, leveraging pixelwise features extracted by convolutional neural networks (CNNs) to explore correlations between spectral pixels. Convolutions are performed primarily in spectral, such as 1-D CNNs [8], [9], 1-D GANs [10], and recurrent neural network (RNN) models [11]. *Spatial feature networks* focus on the spatial features of HSI cubes, integrating the spatial information contained in the neighborhood region of the pixel into classifiers to enhance classification performance. Representative methods in this category include off-the-shelf CNNs [12], [13] for deep spatial features [14] and fully convolutional networks (FCNs) [15], [16] for deep multiscale spatial features. *Spectral–spatial feature networks* extract joint spectral–spatial features, primarily employing the following three approaches.

1) *Preprocessing-Based Networks:* These networks first fuse low-level spectral and spatial features to obtain spectral–spatial features, which are then fed into subsequent deep networks to learn high-level joint spectral–spatial characteristics.
2) *Postprocessing-Based Networks:* These networks first use two separate deep networks to extract spectral and spatial features, then fuse them in the fully connected layers to generate joint spectral–spatial features.
3) *Integrated Networks:* Unlike the above two approaches, these networks use 3-D CNNs to directly extract joint spectral–spatial features from the raw data.

However, the usability of the aforementioned algorithms is entirely dependent on an overly simplistic assumption that the training data [i.e., source domain (SD)] and test data [i.e., target domain (TD)] are independent and identically distributed (i.i.d.). In practice, out-of-distribution (OOD) data is far more common. Models trained solely on SD data overlook the domain shift issue [17], [18], [19], leading to a significant drop in performance on the TD, which severely hinders the large-scale and practical deployment of deep learning models [20], [21]. Numerous studies [22], [23] have shown that the success of traditional deep learning-based HSI classification models is largely driven by supervised learning, heavily relying on the i.i.d. assumption.

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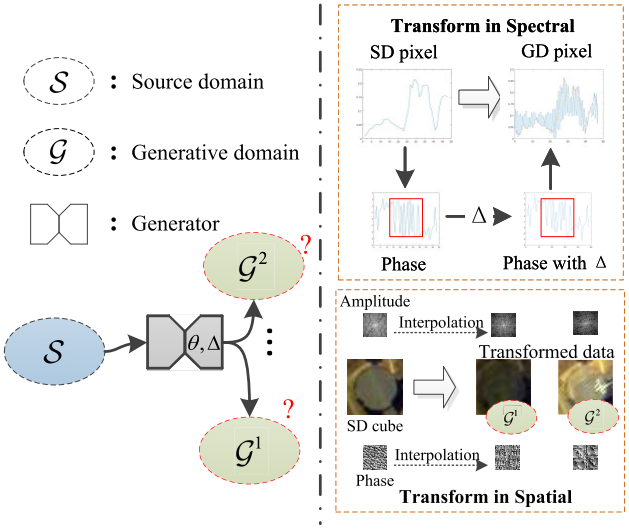


Fig. 1. Main idea. The novel problem setting of FSSG, which aims to design a single-SD generation model based on the principles of semantic consistency and domain information diversification such that it can directly generalize to completely unseen TDs. The main idea to tackle FSSG is by sharing class information in the frequency space of each HSI pixel and learning domain shifts (style shift and land-cover shift) at each HSI cube.

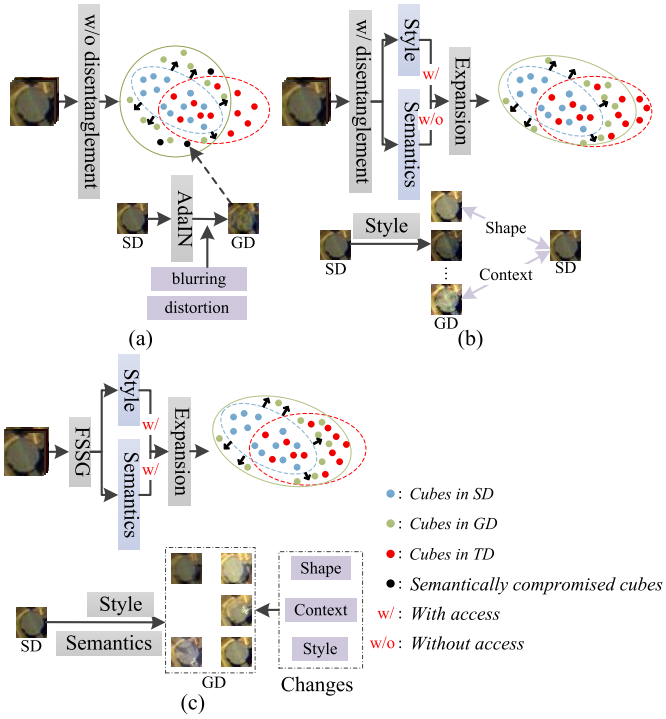


Fig. 2. Comparisons of our proposed FSSG against existing GD-based cross-scene HSI classification algorithms.(a) Illustration of the method without disentanglement. (b) Illustration of the method without context semantic expansion. (c) Illustration of the proposed FSSG.

In recent years, extensive research has been conducted on handling unseen scenes, and numerous cross-scene image classification algorithms have been proposed to address domain shifts in cross-scene HSI classification. The mainstream approach is domain adaptation (DA), which can be broadly divided into discrepancy-based methods and adversarial methods [24]. *Discrepancy-based methods* [25], [26], [27] address domain shifts by matching the marginal or conditional distributions between different domains, with the maximum mean discrepancy (MMD) being a commonly used metric. *Adversarial-based methods* learn correlated and domain-invariant feature representations through adversarial learning. These methods are primarily based on generative adversarial networks (GANs) [28], which consist of a generator model that generates samples similar to those from the SD and a corresponding discriminator [29], [30], [31]. By confusing the discriminator, domain-invariant features are learned.

Cross-scene classification based on domain adaptation (DA) has achieved impressive results; however, it relies on a strong assumption that target data is accessible. On the contrary, in many applications, target data is difficult to obtain or even unseen. For example, in agricultural monitoring, it is impractical to collect data covering all monitoring scenarios under all weather conditions. On the other hand, when the platform carries low computing resources, even if the target data is collected, it cannot be applied to training. To address this challenge, DG is introduced to use data from one or multiple related but distinct SDs to train a model that can generalize indiscriminately to any unseen OOD TG.

Currently, DG-based methods focus on learning domain-invariant features from a single SD [32], [33], [34], [35], [36], [37] (domain expansion networks). As shown in the left-hand side of Fig. 1, the main idea is to perturb the SD to generate pseudo-SDs (similar but distinct from the SD) and then extract universal features from both the source and generated domains for domain-invariant representation learning. For example, Li et al. [33] proposed the progressive domain expansion network (PDEN), where multiple domains are generated to simulate photometric and geometric transformations in the unseen domains. It learns domain-invariant features by minimizing the distance between samples from different domains with the same semantics using contrastive learning. Cheng et al. [35] leveraged adversarial learning and Bayesian neural networks to augment SD, expanding the coverage of the training domain and enhancing the generalization ability of the classifier. Zheng et al. [36] proposed adversarial learning with semantic transformation (AdvST), which augments SD with semantic transformation and uses the augmented data to learn a robust model. Traditional domain expansion networks designed for natural images explore domain-invariant representation only from the spatial dimension. However, domain shifts in HSIs are driven by both spatial texture variations and spectral drift, especially spectral drift. Therefore, recent studies [38], [39] have incorporated spectral information into traditional domain expansion approaches to provide new insights for DG-based cross-scene HSI classification and achieve a surprising classification accuracy. In particular, researchers [40], [41] suggest that expanding the distribution of SD should not solely focus on randomizing the sample styles, but also establish reasonable support. This further improves the classification accuracy.

Despite their promising success in OOD performance, existing DG-based HSI classification methods usually adopt only style randomization to generate generative domain (GD) samples, which introduces some inevitable problems.

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As shown in Fig. 2(a), early DG-based methods lack the ability for disentanglement, and directly randomize the entire HSI cube. Unfortunately, such a paradigm inevitably creates undesirable artifacts (e.g., blurring or distortion), which leads to semantically compromised samples. As shown in Fig. 2(b), current methods ignore the domain shift problem caused by the variance of spatial land-cover distribution, which results in the generated context semantics in GD being fixed and consistent with SD. Even worse, the SD-specific patterns are overly emphasized during the training stage, but they may not be related to the classes in TD. This brings about limited generalizability.

To address the aforementioned issues, a Fourier-based spectral–spatial generator (FSSG) is proposed for domain-invariant representation learning in cross-scene HSI classification. It consists of a Fourier disentanglement mechanism, a spectral expansion (SpeE) module, and a spatial expansion (SpaE) module. Specifically, the Fourier disentanglement mechanism decouples the styles and semantics from pixelwise in SpeE and regionwise in SpaE respectively, which is an effective tool to avoid semantically compromised samples. In SpeE, the style information (phase) is transmitted across pixels in a privacy-protecting way, i.e., the semantic information (amplitude) of each pixel in SD is shared in GD. In SpaE, an effective continuous frequency space interpolation strategy is designed to transmit the style (amplitude) and semantic information (phase) across cubes, which enables GD to simulate interdomain gaps with SD not only from style but also from context semantics. To avoid overemphasis on SD-specific patterns during the training stage, an adversarial training based on a coarse-to-fine paradigm is proposed, where a relaxation procedure is introduced to apply semantic consistency constraints on each pixel. This procedure facilitates the HSI cube to gain robust context semantics, and thus more general classification patterns are learned.

The main contributions of this article are summarized as follows.

1) A novel FSSG is designed to generate robust GDs with both style and context semantic variations to simulate the interdomain gap between the unseen domain and SD. To the best of our knowledge, this is the first work to explore context semantic variations in DG-based HSI classification tasks.
2) A Fourier spectral disentanglement theory is proposed, which theoretically proves that amplitude and phase components in the HSI spectral dimension encode semantic invariance and style variability, respectively. This provides principled guidance for cross-domain style perturbation while preserving semantics.
3) The continuous interpolation mechanism can smoothly bridge different distributions to further enrich the diversity of GD distribution. This also promotes local learning, improving the likelihood of achieving domain invariance in unseen dense distribution space.
4) To avoid overemphasis on SD-specific patterns, a relaxation procedure is introduced into the adversarial training based on the coarse-to-fine paradigm, which facilitates the HSI cube to gain more robust context semantics.

The rest of this article is organized as follows. Section II briefly describes the relevant methods of DG-based HSI classification and Fourier-Based augmentation. In Section III, we carefully describe the preliminaries and implementation details of the proposed method FSSG. Then, we introduce FSSG for domain-invariant representation learning based on two baselines in Section IV. Extensive experiments with competitive results are shown in Section V. Finally, Section VI summarizes the work of this article.

II. Related Work

A. DG-Based HSI Classification

In cross-scene HSI classification tasks, the target scenes often have significant uncertainty and real-time performance is commonly required in practice. To overcome these challenges, some tentative studies [38], [39], [40], [41], [42], [43], [44], [45] have been conducted on cross-scene HSI classification based on the domain generalization (DG) [46] framework. These methods can be broadly divided into two categories: feature alignment and domain adversarial learning. *Feature alignment* aims to make the feature distributions of different sample points identical or similar through specific methods. Based on this theory, Gao et al. [42] proposed a label-indicate-conditional-alignment (LiCa) method, focusing on aligning the spectral conditional distribution of different domains, addressing the issue of spectral heterogeneity. Zhang et al. [43] used information such as category, color, and position to design text representations and then employed a contrastive learning-based visual-language alignment module to extract domain-invariant features. Although considerable results were achieved, the domain-invariant properties revealed by positional relations and colors have limited generalization ability.

Recent research has increasingly focused on *domain adversarial learning*. Zhang et al. [38] developed a single-SD expansion network (SDEnet) that uses adaptive instance normalization (AdaIN) to generate expanded domains from both spatial and spectral perspectives, ensuring domain invariance of the generated samples with morphological mechanisms. Supervised contrastive learning is used to design adversarial loss for sample rationality and contrastive loss for domain-invariant representation learning. Chen et al. [39] proposed a spectral–spatial adversarial multidomain synthesis network (S^2AMSnet), which increases model diversity by expanding the different distributions of the SD, and further enhances diversity with a multidomain mixing module. To ensure semantic consistency between GD and SD in terms of class categories, a multiscale mutual information regularization network is designed, which maximizes the mutual information between GD patches and original spectral patches at different spatial scales to avoid semantic deviation. In the domain-invariant representation stage, a semantic consistency discriminator is developed to learn domain-invariant features from the GD. Peng et al. [40] proposed a progressive feature disentanglement network to

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decompose the original features into domain-invariant features and domain-specific features from three stages: shallow, deep, and semantic features, the obtained domain-invariant features are applied to the classification task. Qin et al. [41] proposed frequency disentanglement and data geometry network (FDGNet), which generates semantically consistent and diverse GDs by transferring high-frequency information and resampling low-frequency information in frequency space. Additionally, data geometry was used to promote the rationality of GD.

The aforementioned methods utilize adversarial generation theory to design domain-invariant representation learning for cross-scene HSI classification, which is both innovative and insightful. However, these methods strongly rely on the assumption that the semantics of SD are still valid in TD. Evidently, this assumption is challenging to achieve in practice.

B. Fourier-Based Augmentation

The Fourier amplitude contains rich low-level statistical features, while the phase reveals higher level semantic features (visual structure). As a result, many studies [47], [48], [49], [50], [51], [52] have been guided by Fourier transforms to design generalized representation learning frameworks. These methods assume that the statistical features contained in the Fourier amplitude correspond to domain-specific information, while the phase represents domain-invariant information. Based on it, Yang et al. [47], Iizuka et al. [50], and Lu et al. [51] reduce the perceptual gap between the source and TDs by exchanging the amplitude between the source and TDs, thereby enhancing the generalization ability in the TD. Xu et al. [48] linearly interpolate the amplitudes of images from different domains to generate new styles, while keeping the phase unchanged to highlight domain-invariant features. Liu et al. [49] developed a continuous frequency–space interpolation mechanism to transfer distribution information (controlled by amplitude) across multiple SDs, using distributions from multiple SDs to solve the DG problem. Guo et al. [52] provide a new clue from another perspective. They revealed that low-frequency regions of the signal represent low-level statistical information, while high-frequency regions represent semantic information.

For hyperspectral signals, Hong et al. [53] introduced invariant attribute profiles (IAPs); IAPs leverage spatial–frequency joint features for HSI classification, it primarily targets static feature extraction and lacks mechanisms for dynamic DG. Zhao et al. [54] provided fractional Fourier image transformer (FrIT), this method integrates fractional Fourier transforms to capture both global and local features in multimodal HSI-LiDAR data. However, FrIT focuses on multimodal fusion rather than addressing domain shifts in single-source DG scenarios. Our work differs by explicitly disentangling spectral–spatial features to simulate both style and context semantic variations in unseen domains. The mechanism enabling this achievement is frequency disentanglement:

Given an input pixel/cube $x \in \mathbb{R}^{1\times1\times C}$ or $X \in \mathbb{R}^{H\times W\times C}$, its frequency space signal is obtained via a Fourier transform as follows: $\mathcal{F}(x)(u) = \sum_{c=0}^{C-1} x(c)e^{-j2\pi((c/C)(u))}$ (1-D Fourier transform for pixel) or $\mathcal{F}(\widehat{X})(u,v,c) = \sum_{h=0}^{H-1}\sum_{w=0}^{W-1}\widehat{X}(h,w,c)e^{-j2\pi((h/H)u+(w/W)v)}$ (2-D Fourier transform [55] for cube). The amplitude $\mathcal{A} = |\mathcal{F}(x)|$ and phase $\mathcal{P} = \angle\mathcal{F}(x)$ are then used to represent style and semantic information, respectively. During the perturbation process, we use masks $\mathcal{M}_a$ and $\mathcal{M}_p$ (which are introduced in detail in Sections III-B and III-C) to further constrain the consistency of semantics, i.e., $\mathcal{A} = \mathcal{A} \odot \mathcal{M}_a$ and phase $\mathcal{P} = \mathcal{P} \odot \mathcal{M}_p$.

III. Method

This section starts with the formulation for single-SD generation and its challenges in the cross-scene HSI classification scenario. Then, the proposed method FSSG is described to explicitly address these challenges. An overview of FSSG is shown in Fig. 3.

A. Single-SD Generation

1) Preliminaries: In the single-source DG model, let $(\mathcal{X}, \mathcal{Y})$ denote the joint space of a task, where $\mathcal{X}$ represents the images and $\mathcal{Y}$ represents the corresponding labels. $\mathcal{S} = \{(X_i, y_i)\}_{i=1}^{N}$ is considered as a known SD distribution $(\mathcal{X}^{\mathcal{S}}, \mathcal{Y})$, where X_i is an HSI cube consisting of the central pixel x_i and its adjacent pixels. The objective of a single-source DG model is to learn a model $f_\theta{:}\mathcal{X} \to \mathcal{Y}$ using only the data from the SD, such that the model can still generalize effectively to an entirely unseen test domain $\mathcal{T}$. These methods primarily enhance the ability of the model to generalize to unseen data by expanding the SD distribution. Specifically, GD sets with different distributions are created by perturbing the SD data, denoted as $\mathcal{G} = (\mathcal{G}^1, \mathcal{G}^2, \ldots, \mathcal{G}^J)$. To ensure the efficacy and safety of data mechanisms, the perturbation only induces distributional changes in the style features, while the spatial properties that represent the basic class semantics should remain consistent with the SD. This article utilizes the Fourier transform in the frequency space to explore low-level styles and high-level semantics, designing a generator that selectively perturbs these two components according to task requirements. Notably, the proposed FSSG can be flexibly integrated into other single-SD extension networks.

2) Challenges: The objective of generalization to unseen domains requires the generator to produce a diversified set of pseudo-SDs to simulate any possible distribution, enabling the learning of domain-invariant representations in the latent space. However, this goal presents specific challenges in the domain expansion setup for cross-scene HSI classification. First, unlike natural images, domain shifts in HSI images involve not only spatial texture changes but also spectral drift. This requires the generator to consider data augmentation from both spatial and spectral. Second, domain shifts in HSI cubes also include variances in the context semantics, which poses a challenge to previous domain expansion methods that often simply consider the context semantics in SD to be shareable domain-invariant information [shown in Fig. 2(b)].

B. Domain Expansion in Spectral

To accommodate the unique data acquisition characteristics of HSI, this work expands the distribution of SD from

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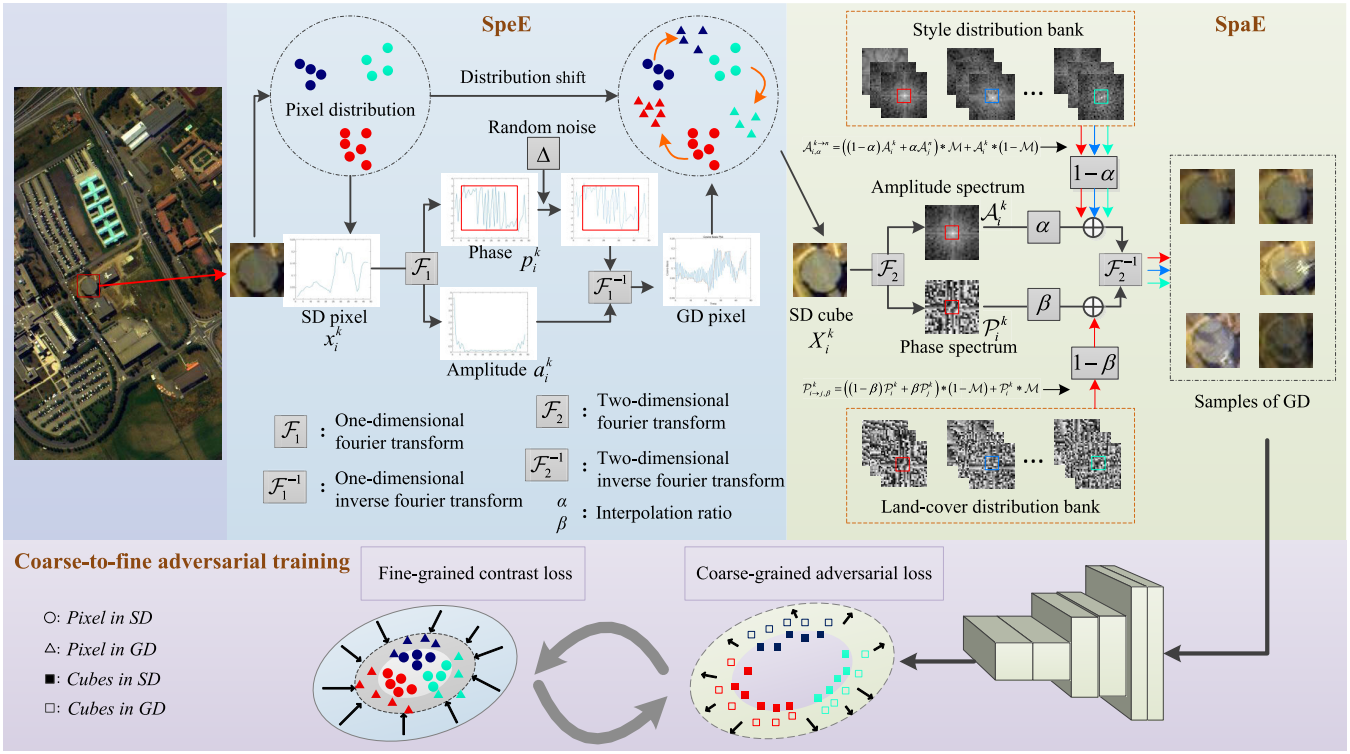


Fig. 3. Schematic of S2AMSnet for cross-scene HSI classification. In SpeE, each pixel of GD is generated by transmitting the phase in frequency space and its semantics are kept consistent with SD by sharing the amplitude. In SpaE, the styles and context semantics are exchanged across cubes using a continuous interpolation mechanism in the frequency space, which enables the generated domain to have diverse styles and context semantics, thereby enhancing its generalization ability to unseen domains. To ensure that HSI cubes of the same class maintain consistent class semantics, the continuous interpolation mechanism on the phase is only performed in the frequency space of innerclass cubes.

both spectral and spatial perspectives. Given that the spectral dimension in HSI data contains essential class-specific (semantic) information, which is crucial for classification tasks, it is especially important to avoid altering class information during perturbations. This article proposes leveraging inherent information in the frequency space to separate class-specific information from the original data, allowing perturbations to occur only in domain-specific information unrelated to the class. This ensures that the semantics of each pixel in GD remain consistent with the semantics of SD.

Specifically, given an HSI pixel $x_i^k \in \mathbb{R}^{1\times1\times C}$ (where C represents the number of spectral bands, and k denotes the kth class), its frequency space signal is obtained via a 1-D Fourier transform as follows:

$$\mathcal{F}_1\left(x_i^k\right)(u) = \sum_{c=0}^{C-1} x_i(c) e^{-j2\pi\left(\frac{c}{C}(u)\right)}. \tag{1}$$

This frequency space signal $\mathcal{F}_1(x_i^k)$ can be further decomposed into an amplitude spectrum $a_i^k \in \mathbb{R}^{1\times1\times C}$ representing the semantic and a phase spectrum $p_i^k \in \mathbb{R}^{1\times1\times C}$ reflecting style-specific information, as shown in Fig. 4. To implement consistent perturbations to style information by class, a style information bank $p = [p^1, p^2, \ldots, p^K]$ is constructed, where each element $p^k = \{p_i^k\}_{i=1}^{N^k}$ (N^k denotes the number of pixels belonging to the kth class) contains the phase spectrum of all HSI pixels from the kth class, representing the style information of all pixels in the k class in $\mathcal{S}$. Perturbations are then applied to this bank to generate style information for representing $\mathcal{G}$.

Next, a random perturbation mechanism is designed in the frequency space to alter the inherent domain distribution information specific to SD, aiming to approximate the distribution of unseen data. As shown on the left side of Fig. 3, given a single pixel x_i^k in the SD kth class, its phase spectrum can be perturbed to generate new domain distribution information while keeping the amplitude spectrum fixed to preserve the semantic information (shown in Fig. 4). As a result, the method presented in this article can generate HSI pixels with different domain distribution information, thus expanding the distribution characteristics of the training samples. In this manner, FSSG can construct diversified GD data in the spectral dimension to improve the effectiveness of the subsequent domain-invariant features extracted. Formally, the concept of differential privacy is introduced, and noise conforming to a Laplace distribution is added to each sample in the style information bank to generate a new phase spectrum, which is represented as follows:

$$p_{i,\varepsilon}^{k\to k+\Delta} = \left(p_i^k + (L_1, L_2, \ldots, L_C)\right) * m + p_i^k * (1-m) \tag{2}$$

where $L_i \sim \text{Lap}(\varepsilon) = 1/(2\varepsilon)\exp(-(|L|)/\varepsilon)$ is an i.i.d. random variable, ε represents the perturbation level of the noise on the original phase spectrum, and $m = \mathbb{1}_{c\in[-\lambda C:\lambda C]}$ is a binary mask that controls the scale of low-frequency component within the phase spectrum to be perturbed, where

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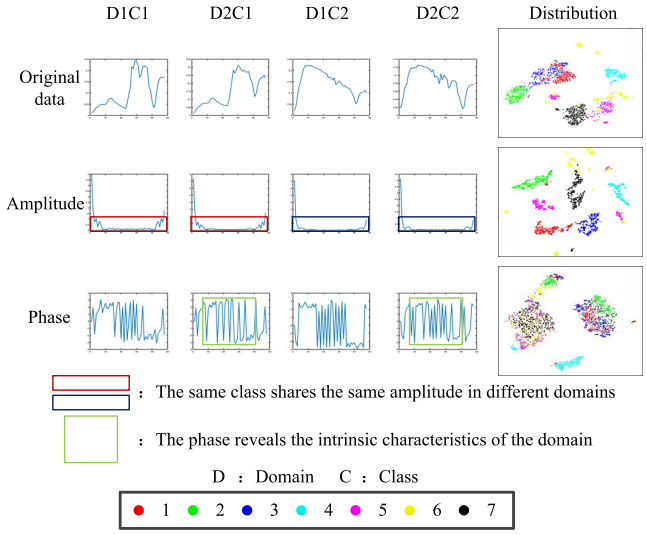


Fig. 4. Characteristics of spectral pixels in frequency. Note: "D" and "C" denote domain and class, respectively. For example: "D1C1" represents Class 1 samples in Domain 1. Explanation: Samples in the same class from different domains share the amplitude at the high-frequency component after decentralization, and the information contained in amplitude has a strong clustering effect. The difference between them is mainly reflected in the phase spectrum at the low-frequency component (note that the amplitude spectrum at this component cannot describe the interdomain gaps). Therefore, this work regards the amplitude as the semantics and the phase at the low-frequency component as the domain-specific information.

$m = 1$ in the central region and $m = 0$ in other regions [49]. Once the perturbed phase spectrum $p_{i,\varepsilon}^{k\to k+\Delta}$ is obtained, it is combined with the original amplitude spectrum to generate a spectral pixel of GD through a 1-D inverse Fourier transform, as follows:

$$x_{i,\varepsilon}^{k\to k+\Delta} = \mathcal{F}_1^{-1}\left(a_i^k, p_{i,\varepsilon}^{k\to k+\Delta}\right) \tag{3}$$

where the generated pixel $x_{i,\varepsilon}^{k\to k+\Delta}$ retains the original semantic information of x_i^k while acquiring new style information. Throughout the perturbation process, the upper bound of semantic consistency error is constrained (e.g., $E_p/E_x = (\|\hat{x} - x\|_2)/(\|x\|_2) \le (\varepsilon\sqrt{C})/(\sum_u a(u))$), and the process itself ensures linear control over output variations (see Appendices A and B for the proof). For convenience, the result is denoted as $\widehat{x}_i^k$, yielding the preliminary $\mathcal{G} = (\widehat{x}(i), y_i)i = 1^N$. Moreover, this method does not violate ethical standards, as it only utilizes SD data throughout the process and generates new data using random perturbations unrelated to TD.

C. Domain Expansion in Spatial

Although spectral information contains important class-related information, spatial neighborhood relationships are also crucial for HSI data classification. Therefore, DG in HSI classification should consider domain shifts in the spatial dimension. This issue arises because the context semantics of SD become invalid in TD due to variances in land-cover distribution, which have been overlooked by previous HSI classification methods based on single-SD expansion, limiting the generalizability of the model. This work generates GD samples from both styles and context semantics. In detail, by utilizing the regionwise Fourier disentanglement mechanism, the style and semantic information of the original HSI cube are decoupled, and each is then perturbed to generate new samples. In this process, GD is defined as follows: $\mathcal{G} = (\widehat{X}i, y_i)i = 1^N$.

Specifically, given an HSI cube $\widehat{X}_i^k \in \mathbb{R}^{H\times W\times C}$ ((H, W), C representing the height, width, and number of bands, respectively), its frequency space signal is obtained via a 2-D fast Fourier transform as follows:

$$\mathcal{F}_2\left(\widehat{X}_i^k\right)(u, v, c) = \sum_{h=0}^{H-1}\sum_{w=0}^{W-1} \widehat{X}_i^k(h, w, c)e^{-j2\pi\left(\frac{h}{H}u + \frac{w}{W}v\right)}. \tag{4}$$

This frequency space signal $\mathcal{F}_2(\widehat{X}_i^k)$ can be further decomposed into the amplitude spectrum $\mathcal{A}_i^k \in \mathbb{R}^{H\times W\times C}$ and the phase spectrum $\mathcal{P}_i^k \in \mathbb{R}^{H\times W\times C}$, representing the low-level distribution (e.g., style) and high-level semantic (e.g., context semantic) of the HSI cube, respectively. As DG theory requires that only SD data participate in training, this work introduces cross-cube style perturbations, whereby the styles of different samples in a minibatch are exchanged to expand the spatial styles. A spatial style bank $\mathcal{A} = [\mathcal{A}^1, \mathcal{A}^2, \ldots, \mathcal{A}^K]$ is first constructed, where each element $\mathcal{A}^k = \{\mathcal{A}_i^k\}_{i=1}^{N^k}$ contains all HSI cubes from the kth class, representing the styles. All HSI cubes can access this bank.

Next, a continuous interpolation mechanism is designed within a continuous space to transfer style information between different samples and expand the distribution of SD. As shown on the right side of Fig. 3, given an HSI cube $\widehat{X}_i^k$ from the kth class in the SD, a portion of the low-frequency components of the amplitude can be replaced with those in bank $\mathcal{A}$. This allows the model to acquire a style different from that of the SD. More importantly, continuous interpolation is performed between the amplitude of the current class and the transferred amplitude spectrum of other classes. This dedicated dense-space method, which smoothly changes the distribution, enriches the style distribution of the GD. Formally, the amplitude $\mathcal{A}_j^n$ is randomly sampled from the style bank, and interpolation is performed between $\mathcal{A}_i^k$ and $\mathcal{A}_j^n$ to synthesize a new amplitude spectrum. Let $\mathcal{M} = \mathbb{1}_{(h,w)\in[-\lambda H:\lambda H,-\lambda W:\lambda W]}$ be a binary mask with the same function as m in Section III-B. The interpolation rate between $\mathcal{A}_i^k$ and $\mathcal{A}_j^n$ is denoted as α, and the new amplitude spectrum can be represented as follows:

$$\mathcal{A}_{i,\alpha}^{k\to n} = \left((1-\alpha)\mathcal{A}_i^k + \alpha\mathcal{A}_j^n\right) * \mathcal{M} + \mathcal{A}_i^k * (1-\mathcal{M}). \tag{5}$$

To generate GD with different context semantics, a semantic bank $\mathcal{P} = [\mathcal{P}^1, \mathcal{P}^2, \ldots, \mathcal{P}^K]$ is also constructed, where each element $\mathcal{P}^k = \{\mathcal{P}_i^k\}_{i=1}^{N^k}$ contains all HSI cubes from the kth class, representing the semantic information. *Note that*, privacy-preserving mechanisms are implemented based on categorical attributes, meaning information within this bank is exclusively shared intraclass while remaining restricted from interclass access. A schematic of this architecture is illustrated in Fig. 5. Given $\widehat{X}_i^k$, a portion of the low-frequency components of the phase can be replaced with those from the bank $\mathcal{P}$. Let β denote the interpolation rate between $\mathcal{P}_i^k$ and $\mathcal{P}_j^k$. The new phase is then represented as follows:

$$\mathcal{P}_{i\to j,\beta}^k = \left((1-\beta)\mathcal{P}_i^k + \beta\mathcal{P}_j^k\right) * (1-\mathcal{M}) + \mathcal{P}_i^k * \mathcal{M}. \tag{6}$$

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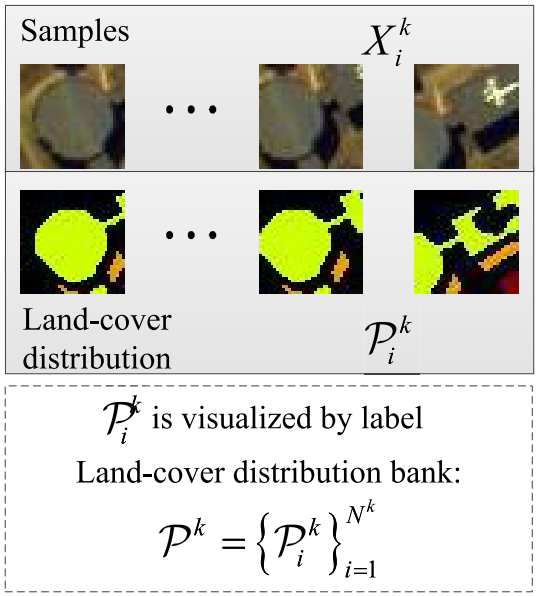


Fig. 5. Schematic of land-cover distribution.

After obtaining the interpolated amplitude $\mathcal{A}_{i,\alpha}^{k\to n}$ and phase $\mathcal{P}_{i\to j,\beta}^{k}$, both are combined, and a 2-D inverse Fourier transform is applied to generate the transformed HSI cube as follows:

$$\widehat{X}_{i,\alpha,\beta}^{k\to n} = \mathcal{F}_2^{-1}\left(\mathcal{A}_{i,\alpha}^{k\to n}, \mathcal{P}_{i\to j,\beta}^{k}\right) \tag{7}$$

where the resulting HSI cube $\widehat{X}_{i,\alpha,\beta}^{k\to n}$ retains the class attribute of the SD while changing the styles and context semantics. During the implementation, the interpolation ratio α and β is dynamically sampled from [0.0, 1.0] to generate GD samples in the continuous distribution space. For convenience, the transformed HSI cube is denoted as $\hat{X}_i^k$, resulting in the final GD: $\mathcal{G} = \{(\hat{X}_i, y_i)\}_{i=1}^N$. It is worth noting that perturbing the high-level context semantics, which represents the land-cover distribution, alters the inherent land-cover distribution of the SD, allowing the model to learn unseen spatial neighborhood relations. This aligns with the needs of practical applications.

D. FSSG Optimization

The expanded distribution constructed in the SD provides the material for learning generalizable classification features. In this section, to address the challenges of model generalizability in cross-scene HSI classification tasks, we propose a carefully designed coarse-to-fine adversarial learning strategy that ensures the robustness of GD.

As mentioned in Section III-A, the context semantics of SD samples may not cover those of TD, which leads to overemphasis on SD-specific patterns during the training stage, but they may not be related to the classes in TD. To solve this problem, a relaxation procedure is introduced to apply semantic consistency constraints to each pixel. This procedure enables the HSI cube to gain more robust context semantics. To achieve this goal, a pixelwise semantic consistency loss called *fine-grained contrast loss* is introduced. Meanwhile, to ensure the effectiveness of GD, a cubewise adversarial loss is proposed to encourage FSSG to generate samples outside the SD distribution, which is called *coarse-grained adversarial loss*.

1) Fine-Grained Contrast Loss: As shown in Fig. 3, conceptually, perturbations in SpeE do not alter the class property. However, in SpaE, the continuous interpolation of the amplitude and phase also leads to changes in spectral, which may result in semantic drifts in individual spectral pixels. To ensure the semantic invariance of each pixel, a pixelwise supervised contrastive loss [56] is designed

$$\mathcal{L}_{\text{con}}^{G} = \sum_{i}^{N} \frac{-1}{|P(i)|} \sum_{j\in P(i)} \log \frac{\exp\left(\mathbf{x}_i \cdot \mathbf{x}_j^{+}/\tau\right)}{\sum_{m\in A(i)} \exp\left(\mathbf{x}_i \cdot \mathbf{x}_m^{-}/\tau\right)} \tag{8}$$

where, for spectral feature $\mathbf{x}_i$ of each pixel output by FSSG in minibatch, $\mathbf{x}_j^{+} \in P(i)$ represents a member from the positive sample set, $\mathbf{x}_m^{-} \in A(i)$ represents a member from the negative samples set, and $|P(i)|$ represents the number of positive samples. During the optimization process, the positive sample set $P(i)$ is composed of all same-class pixels from the SD and GD, while pixels from different classes are placed in the negative sample set $A(i)$. Semantic consistency $\mathcal{L}_{\text{con}}^{G}$ is enforced by optimizing the loss to ensure that each pixel in the GD maintains semantic consistency with the SD, preventing the class property of the pixel from being changed.

2) Coarse-Grained Adversarial Loss: To ensure the effectiveness of the GD, i.e., the HSI cubes in the GD should deviate from the SD to simulate the distribution of the unseen domain, the cubewise adversarial loss is introduced. In contrast to the semantic consistency constraint, the samples of kth class from SD are treated as positive samples, and the corresponding samples of the same class from GD are treated as negative samples

$$\mathcal{L}_{\text{adv}} = \sum_{k} \sum_{i}^{N^k} \frac{-1}{|P_k(i)|} \sum_{j\in P_k(i)} \log \frac{\exp\left(\mathbf{X}_i \cdot \mathbf{X}_j^{+}/\tau\right)}{\sum_{m\in A_k(i)} \exp\left(\mathbf{X}_i \cdot \mathbf{X}_m^{-}/\tau\right)} \tag{9}$$

where, for spatial feature $\mathbf{X}_i$ of each cube output by FSSG in a minibatch, $P_k(i)$ and $A_k(i)$ represent the sets of positive and negative samples, respectively, with supervision labels set to 1 and 0. During optimization, the GD samples and SD samples from the same class gradually separate, causing the GD to contain diversified styles and robust context semantics that can simulate all possible distributions of the unseen scenarios. In domain-invariant representation learning, $\mathbf{X}_i$ will be replaced by the embedding feature $\mathbf{z}_i$ in the fully connected layer of the domain-invariant feature extractor.

IV. FSSG for Domain-Invariant Representation Learning

In this section, two classic single-source DG methods (SDEnet [38] and S^{2}AMSnet [39]) are selected as baselines to validate the contribution of the proposed FSSG to HSI domain-invariant representation learning. Specifically, we use FSSG to completely replace their generator D, without making any changes to the discriminator G. Both methods use

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cross-entropy as the classification loss

$$\mathcal{L}_{\text{cls}}(\mathcal{X},\mathcal{Y}) = \frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K} -y_i^k \log \mathbf{p}_i^k \tag{10}$$

where $\mathbf{p} = \{\mathbf{p}_i\}_{i=1}^N$ represents the predicted probability of the output, and $\mathbf{p}_i = \Psi_{\text{cls}}(\theta(X_i))$, y_i denotes the class label.

A. FSSG Improving SDEnet

In SDEnet, an intermediate domain (ID) is generated to diversify GD by randomly weighting the SD and GD, denoted as $\mathcal{I} = \{(\tilde{X}_i, y_i)\}_{i=1}^N$.To ensure the reliability of GD, discriminator D is required to correctly predict both GD and ID. Since $\mathcal{S}$, $\mathcal{G}$, and $\mathcal{I}$ share the same label space, their classification losses are as follows: $\mathcal{L}_{\text{cls}}^{\mathcal{S}} = \mathcal{L}_{\text{cls}}(\mathcal{X}^{\mathcal{S}},\mathcal{Y})$ corresponding to SD, $\mathcal{L}_{\text{cls}}^{\mathcal{G}} = \mathcal{L}_{\text{cls}}(\mathcal{X}^{\mathcal{G}},\mathcal{Y})$ corresponding to GD, and $\mathcal{L}_{\text{cls}}^{\mathcal{I}} = \mathcal{L}_{\text{cls}}(\mathcal{X}^{\mathcal{I}},\mathcal{Y})$ corresponding to ID.

To extract domain-invariant features, supervised contrastive learning is introduced in SDEnet to optimize the discriminator D

$$\mathcal{L}_{\text{con}}^{D} = \sum_{i}^{N}\frac{-1}{|P(i)|}\sum_{j\in P(i)}\log\frac{\exp\left(\mathbf{z}_i\cdot\mathbf{z}_j^{+}/\tau\right)}{\sum_{m\in A(i)}\exp\left(\mathbf{z}_i\cdot\mathbf{z}_m^{-}/\tau\right)} \tag{11}$$

where $\mathbf{z}_i$ represents the embedding features in the fully connected layer, and the setup of the positive and negative sample sets is consistent with that in (8). Finally, the overall optimization of the network is achieved by alternately optimizing D and G

$$\begin{aligned}&\min_{D}\mathcal{L}_{\text{cls}}^{\mathcal{S}} + \mathcal{L}_{\text{cls}}^{\mathcal{G}} + \mathcal{L}_{\text{cls}}^{\mathcal{I}} + \lambda_1\mathcal{L}_{\text{con}}^{D}\\ &\min_{G}\mathcal{L}_{\text{cls}}^{\mathcal{G}} + \lambda_2\mathcal{L}_{\text{adv}} + \lambda_1\mathcal{L}_{\text{con}}^{G}\end{aligned} \tag{12}$$

where λ_1 and λ_2 are regularization parameters. In the optimization objective of G, the primary goal of $\mathcal{L}_{\text{cls}}^{\mathcal{G}}$ is to ensure that the generated samples adhere to the correct label space, while $\mathcal{L}_{\text{adv}}$ is used to optimize the generator, encouraging it to produce effective samples, i.e., samples that differ from the SD. $\mathcal{L}_{\text{con}}^{G}$ is designed to ensure that the class semantics of each pixel in the generated sample remain unchanged despite the perturbations.

B. FSSG Improving S^{2}AMSnet

In S^{2}AMSnet, two nonshared parameters D are designed to generate $\mathcal{G}_1$ and $\mathcal{G}_2$ and combine them with $\mathcal{S}$ to generate the IDs $\mathcal{I}_1$ and $\mathcal{I}_2$. Then, a semantic consistency discriminator is designed to learn domain-invariant representations. In particular, the SD data is not directly involved in the model optimization process. The consistency of class semantic features with the SD is ensured by minimizing the classification loss between the GD and ID. The optimization targets are as follows: $\mathcal{L}_{\text{cls}}^{\mathcal{G}_1} = \mathcal{L}_{\text{cls}}(\mathcal{X}^{\mathcal{G}_1},\mathcal{Y})$, $\mathcal{L}_{\text{cls}}^{\mathcal{G}_2} = \mathcal{L}_{\text{cls}}(\mathcal{X}^{\mathcal{G}_2},\mathcal{Y})$, $\mathcal{L}_{\text{cls}}^{\mathcal{I}_1} = \mathcal{L}_{\text{cls}}(\mathcal{X}^{\mathcal{I}_1},\mathcal{Y})$, and $\mathcal{L}_{\text{cls}}^{\mathcal{I}_2} = \mathcal{L}_{\text{cls}}(\mathcal{X}^{\mathcal{I}_2},\mathcal{Y})$.

The semantic consistency discriminator uses KL divergence to measure the discrepancy between the distributions of the GD and ID, and it captures the consistency of class semantics between different domains by minimizing the distribution difference, formulated as follows:

$$\mathcal{L}_{\text{KL}} = \mathcal{L}_{\text{KL}}(\mathcal{G}_1,\mathcal{I}_2) + \mathcal{L}_{\text{KL}}(\mathcal{G}_2,\mathcal{I}_1). \tag{13}$$

The overall loss of the semantic consistency discriminator is

$$\min_{D}\mathcal{L}_{\text{cls}}^{\mathcal{G}_1} + \mathcal{L}_{\text{cls}}^{\mathcal{G}_2} + \mathcal{L}_{\text{cls}}^{\mathcal{I}_1} + \mathcal{L}_{\text{cls}}^{\mathcal{I}_2} + \mathcal{L}_{\text{KL}}. \tag{14}$$

For generator optimization, S^{2}AMSnet maximizes $\mathcal{L}_{\text{KL}}$, which encourages the GD to deviate from the SD data. Additionally, by maximizing the multiscale mutual information between the SD and GD, the generated samples are ensured not to deviate semantically from the SD. For a fair comparison, $\mathcal{L}_{\text{KL}}$ is still used to optimize the generator; however, the multiscale mutual information is replaced by (8). Since this is a maximization optimization problem, (8) is adjusted as follows:

$$\mathcal{L}_{\text{con}} = 1 - \mathcal{L}_{\text{con}}^{G}. \tag{15}$$

The overall loss of FSSG in this method is

$$\max_{G}\mathcal{L}_{\text{con}} + \mathcal{L}_{\text{KL}} \tag{16}$$

where $\mathcal{L}_{\text{con}}^{G}$ gradually decreases as $\mathcal{L}_{\text{con}}$ increases, forcing the class semantics of generated samples to become more similar to those from the SD. It is worth mentioning that FSSG imposes semantic consistency constraints on each pixel through a relaxation procedure, which further enhances the robustness of GD samples.

V. Experimental Results and Analysis

A. Experimental Setup

1) Datasets: The superiority of FSSG is validated on three datasets.

1) *Houston Dataset:* This dataset [57] contains two hyperspectral scenes (Houston 2013 and 2018) collected over the University of Houston campus, both acquired by different sensors at different times, covering the same wavelength range (0.38–1.05 μm) but differing in spectral bands (144 versus 48). For homogeneous analysis with identical category distributions, 48-band data from Houston 2013 (349 $\times$ 1905 pixels) and a 209 $\times$ 955 overlapping region were selected. The seven shared land-cover classes with sample statistics are detailed in Table I, visualized through pseudo-color composites and ground-truth maps in Fig. 6.
2) *Pavia Dataset:* This dataset [58] comprises two reflective optics spectrographic image system (ROSIS) sensor-acquired urban scenes: Pavia University (PU) and Pavia Center (PC), both captured by the ROSIS sensor, spanning 0.43–0.86 μm. PU (610 $\times$ 340 pixels, 103 bands) and PC (1096 $\times$ 715 pixels, 102 bands) share seven common classes. To ensure spectral homogeneity, the 103rd band in PU was removed for dimensional consistency with PC. Class distributions are quantified in Table II, supported by pseudo-color imagery and reference maps in Fig. 7.

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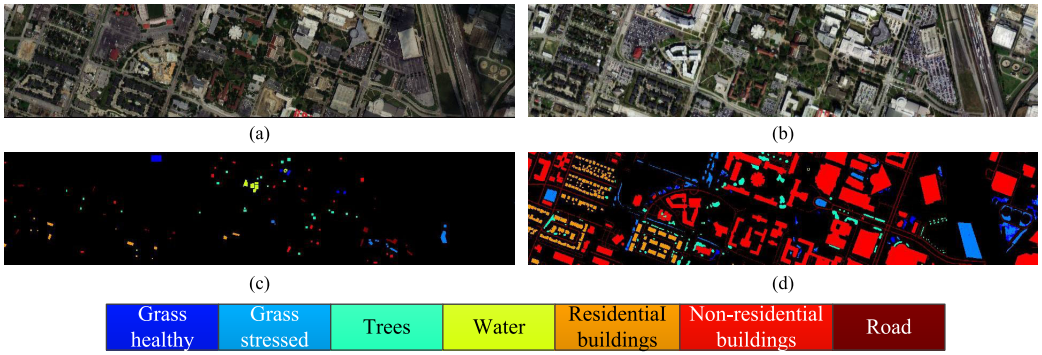


Fig. 6. Pseudo-color images and ground-truth maps of the Houston dataset. (a) and (b) Pseudo-color images of Houston 2013 and Houston 2018, respectively. (c) and (d) Ground-truth maps for Houston 2013 and Houston 2018, respectively.

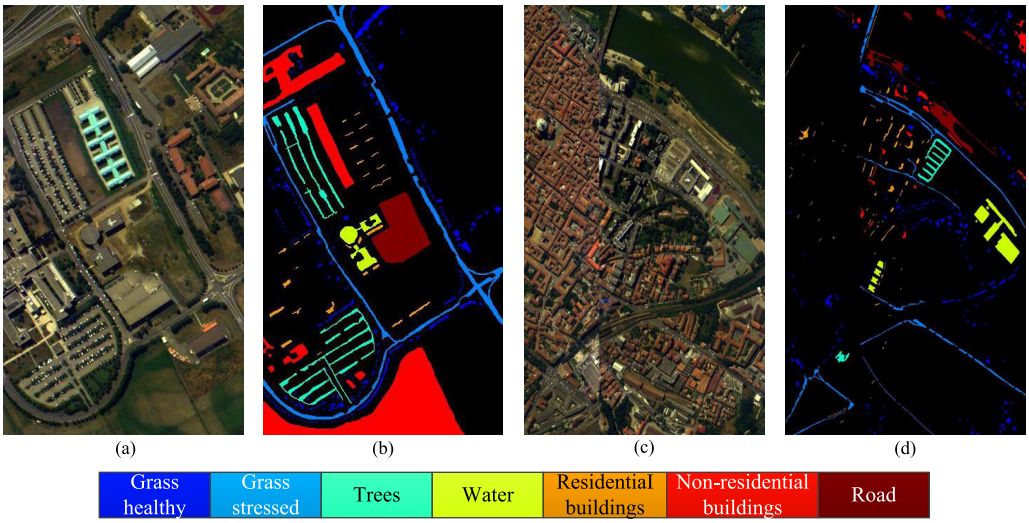


Fig. 7. Pseudo-color images and ground-truth maps of the Pavia dataset. (a) and (c) Pseudo-color images of PU and PC, respectively. (b) and (d) Ground-truth maps for PU and PC, respectively.

TABLE I
NUMBER OF SOURCE AND TARGET SAMPLES OF THE HOUSTON DATASET

Class		Number of Samples	
No.	Name	Houston 2013 (Source)	Houston 2018 (Target)
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365
Total		2530	53200

TABLE II
NUMBER OF SOURCE AND TARGET SAMPLES OF THE PAVIA DATASET

Class		Number of Samples	
No.	Name	PU (Source)	PC (Target)
1	Tree	3064	7598
2	Asphalt	6631	9248
3	Brick	3682	2685
4	Bitumen	1330	7287
5	Shadow	947	2863
6	Meadow	18649	3090
7	Bare soil	5029	6584
Total		39332	39355

3) *HyRANK Dataset:* This dataset [54] was developed under the framework of the International Society for Photogrammetry and Remote Sensing's scientific initiative, using the Hyperion sensor. It consists of two scenes, each containing 176 spectral bands, named Dioni (composed of 250 × 1376 pixels) and Loukia (composed of 249 × 945 pixels). Both scenes share 12 consistent categories. The corresponding sample sizes are shown in Table III, with the pseudo-color images and corresponding ground-truth maps provided in Fig. 8.

4) *WHU-Hi Dataset:* HongHu and HanChuan datasets are selected for validation of FSSG: Both datasets were acquired by the Headwall Nano-Hyperspec sensor with 400–1000 nm spectral coverage. The HongHu scene (940 × 475 pixels) contains 270 spectral bands at 0.043 m spatial resolution, while HanChuan (1217 × 303 pixels) comprises 274 bands with 0.109 m spatial resolution. For experimental consistency, HanChuan's 270-band subset was adopted, targeting three land-cover

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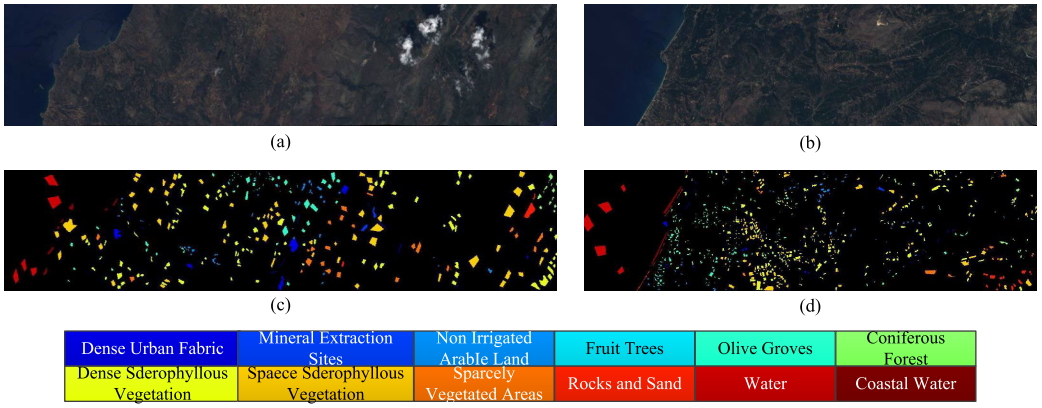


Fig. 8. Pseudo-color images and ground-truth maps of the HyRANK dataset. (a) and (b) Pseudo-color images of Dioni and Loukia, respectively. (c) and (d) Ground-truth maps for Dioni and Loukia, respectively.

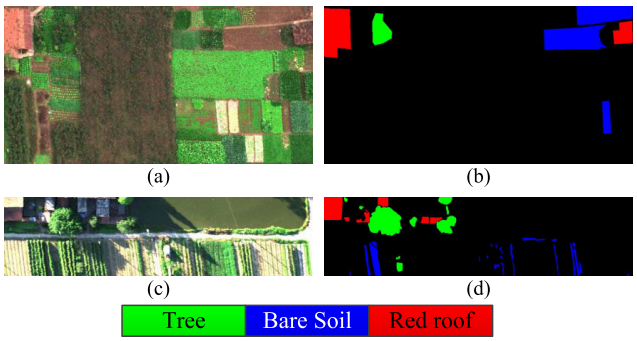


Fig. 9. Pseudo-color images and ground-truth maps of the WHU-Hi dataset. (a) and (c) Pseudo-color images of HongHu and HanChuan, respectively. (b) and (d) Ground-truth maps for HongHu and HanChuan, respectively.

TABLE III
NUMBER OF SOURCE AND TARGET SAMPLES OF THE HYRANK DATASET

Class		Number of Samples	
No.	Name	Dioni (Source)	Loukia (Target)
1	Dense Urban Fabric	1262	206
2	Mineral Extraction Sites	204	54
3	Non Irrigated Arable Land	614	426
4	Fruit Trees	150	79
5	Olive Groves	1768	1107
6	Coniferous Forest	361	422
7	Dense Sderophyllous Vegetation	5035	2996
8	Sparce Sderophyllous Vegetation	6374	2361
9	Sparcely Vegetated Areas	1754	399
10	Rocks and Sand	492	453
11	Water	1612	1393
12	Coastal Water	398	421
Total		20024	10317

classes: tree, red roof, and bare soil. Sample quantities per class are enumerated in Table IV, with Fig. 9 illustrating the corresponding false-color composite and actual terrain distribution.

2) Training: As mentioned in Section IV, the latest two classic single-source DG models, SDEnet and S^{2}AMSnet, were used to validate the superiority of the proposed algorithm. To ensure fairness, all parameter settings in this study are consistent with the two baselines. Specifically, in SDEnet,

TABLE IV
NUMBER OF SOURCE AND TARGET SAMPLES OF THE WHU-HI DATASET

Class		Number of Samples	
No.	Name	HongHu (Source)	HanChuan (Target)
1	Tree	4040	15558
2	Bare soil	21821	9116
3	Red roof	14041	10516
Total		39902	35190

the hyperparameters λ_1 and λ_2 for balancing supervised contrastive learning and adversarial learning were set to $1e+0$ and $1e-1$, respectively. This setting was also applied to the HyRANK dataset. In S^{2}AMSnet, the temperature parameter in the loss function was set to 0.07. For the learning rate, both baselines used a value of $1e-3$. For the input spectral patch size, the two baselines had different settings: in SDEnet, the input spectral patch size was set to 13×13; in S^{2}AMSnet, the input spectral patch sizes for the Houston and WHU-Hi were set to 13×13, and Pavia and HyRANK datasets were set to 9×9 and 7×7, respectively. Unlike SDEnet and S^{2}AMSnet, the training and validation sets split for the Houston dataset were split 80%–20%, while for the other two datasets, the training and validation sets were split equally at 50%–50%. It is noteworthy that no information from the TD was involved during the training phase.

3) Testing: The TD data was fed into the trained classifier, and direct classification was performed without introducing any modifications or additional inference steps. In other words, the projection head was discarded directly.

4) Evaluation Metrics: Performance was evaluated using class accuracy (CA), overall accuracy (OA), and the Kappa coefficient (KC) according to standard evaluation metrics.

5) Experimental Environment: This work was implemented in PyTorch 1.12.1 and trained on an NVIDIA GeForce RTX 3080Ti GPU with 12 GB of RAM. Testing was performed on the same machine.

B. Performance on Cross-Scene HSI Classification

Since FSSG is only a generator proposed to expand the distribution of a single SD. Two authoritative algorithms,

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TABLE V

CLASS-SPECIFIC AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR THE TARGET SCENE HOUSTON 2018 DATA

Class	Classification algorithms							
	PDEN[33]	LDSG[34]	LLUR[45]	DSDG[40]	SDEnet[38]	SDEnet+**FSSG**	S^2MAsnet[39]	S^2MASnet+**FSSG**
1	46.49	10.13	42.68	**60.90**	46.12	38.95	43.09	45.23
2	77.60	62.97	75.30	80.07	77.56	80.32	75.12	**90.14**
3	59.73	60.81	60.34	59.32	42.44	63.59	**69.88**	59.94
4	**100**	81.82	**100**	90.90	**100**	**100**	**100**	**100**
5	49.62	46.65	76.56	**86.25**	57.32	73.97	81.37	71.22
6	84.98	89.22	82.57	87.74	87.76	86.53	92.49	**94.72**
7	**64.21**	44.15	54.85	63.73	60.66	47.54	40.71	37.99
OA (%)	75.40±1.76	74.23±1.49	75.81±1.40	81.98±1.22	77.12±1.72	79.46±1.12 (+2.34)	81.15±0.36	**82.08±0.27** (+0.95)
KC (k)	55.87±2.92	55.42±2.27	60.16±2.19	**70.13±2.28**	60.79±2.05	65.92±1.58 (+4.13)	67.49±1.22	68.28±1.08 (+0.79)

[1] The best and second-best results are highlighted in **bold** and gray

TABLE VI

CLASS-SPECIFIC AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR THE TARGET SCENE PC DATA

Class	Classification algorithms							
	PDEN[33]	LDSG[34]	LLUR[45]	DSDG[40]	SDEnet[38]	SDEnet+**FSSG**	S^2MAsnet[39]	S^2MASnet+**FSSG**
1	85.93	91.09	89.38	80.81	83.57	90.44	85.86	**94.1**
2	88.56	73.51	86.04	89.07	86.51	86.14	**90.6**	88.67
3	61.34	2.23	60.32	**90.14**	64.95	82.27	85.77	76.69
4	**85.49**	71.72	77.4	78.19	85.08	85.29	80.58	81.94
5	87.95	71.04	87.89	85.68	80.3	**98.29**	87.29	93.01
6	79.26	52.12	72.58	**86.96**	77.51	75.57	74.72	64.27
7	64.75	78.13	78.87	80.71	68.65	68.35	74.74	**89.98**
OA (%)	80.27±1.38	70.88±1.55	81.21±1.25	83.72±1.68	80.06±0.62	83.63±1.11 (+3.57)	83.36±1.35	**86.27±0.65** (+2.91)
KC (k)	76.81±1.92	64.06±2.07	77.37±1.50	80.52±2.53	76.07± 0.76	80.41±1.61 (+4.34)	80.04±1.86	**83.44±0.88** (+3.4)

[1] The best and second-best results are highlighted in **bold** and gray

TABLE VII

CLASS-SPECIFIC AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR THE TARGET SCENE LOUKIA DATA

Class	Classification algorithms							
	PDEN[33]	LDSG[34]	LLUR[45]	DSDG[40]	SDEnet[38]	SDEnet+**FSSG**	S^2MAsnet[39]	S^2MASnet+**FSSG**
1	17.48	0	30.29	7.28	28.16	**33.98**	16.5	2.43
2	**38.89**	0	2.78	35.18	5.56	1.85	12.96	0
3	45.54	0.47	46.74	28.17	**64.95**	50.47	57.28	43.9
4	7.59	0	5.06	20.25	12.66	31.65	2.53	**34.18**
5	0.45	0	0.37	10.93	0.99	0.54	0	**73.44**
6	41.94	0	36.54	16.59	**47.63**	31.28	38.15	40.5
7	68.56	78.2	76.37	**84.85**	74.3	76.97	79.81	65.12
8	71.33	64.25	68.42	63.10	61.54	73.06	**83.1**	78.14
9	52.63	15.79	30.78	67.67	**75.69**	37.34	10.28	36.59
10	0	5.74	**17.92**	11.4	0	0	0	2.21
11	**100**	99.86	**100**	**100**	**100**	**100**	**100**	**100**
12	**100**	47.98	99.95	99.29	**100**	**100**	**100**	**100**
OA (%)	60.10±2.86	53.73±2.77	61.52±1.11	63.28±3.89	60.22±0.72	62.45±0.42 (+2.23)	64.51±0.44	**67.53±0.42** (+3.02)
KC (k)	50.43±2.93	41.03±3.11	52.18±1.27	54.06±3.72	50.97± 1.02	53.12±0.43 (+2.15)	55.12±0.47	**60.68±0.73** (+5.56)

[1] The best and second-best results are highlighted in **bold** and gray

SDEnet [38] and S^2MASnet [39], are selected for baselines to evaluate their contribution to HSI domain-invariant representation learning. For further analysis of the generalizability of FSSG, relevant DG-based methods including PDEN [33], LDSG [34], LLUR [45], and DSDG [40] are used for quantitative comparisons.

1) Quantitative Results: Tables V–VII present quantitative comparison results on three public cross-scene HSI datasets.

1) *Contributions of FSSG:* Columns 6–9 of Tables V–VII and columns 2–5 of Table VIII present comparison results of the two baselines (i.e. SDEnet and S^2MASnet) and their improved version by FSSG on four datasets in terms of CA, OA, and KC. It can be objectively found that, by limiting the disturbance mechanism to domain information and exchanging semantics across inner-class HSI cubes, the performance of baseline models consistently improves. Specifically, for the target scene Houston 2018, 2.34/0.95 points gain over SDEnet and S^2MASnet, respectively; for the target scene PC, 3.57/2.91 points gain over SDEnet and S^2MASnet, respectively; for the target scene Loukia, 2.23/3.02 points gain over SDEnet and S^2MASnet, respectively; and for the target scene HanChuan, 2.93/1.18 points gain over SDEnet and S^2MASnet, respectively. KC also saw significant improvements, the highest increase is 4.62/5.56 points gain over SDEnet and S^2MASnet, respectively.

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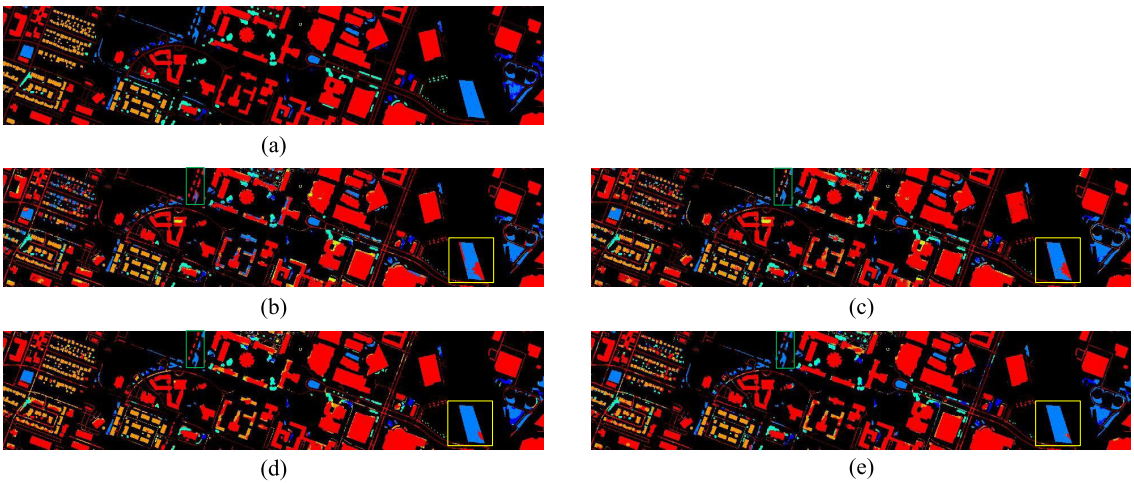


Fig. 10. Classification maps for target scene Houston 2018 dataset obtained using two baselines and corresponding FSSG-improved methods. (a) Ground truth, (b) SDEnet (77.12%), (c) SDEnet + FSSG (79.46%), (d) S2AMSnet (81.15%), and (e) S2AMSnet + FSSG (82.08%).

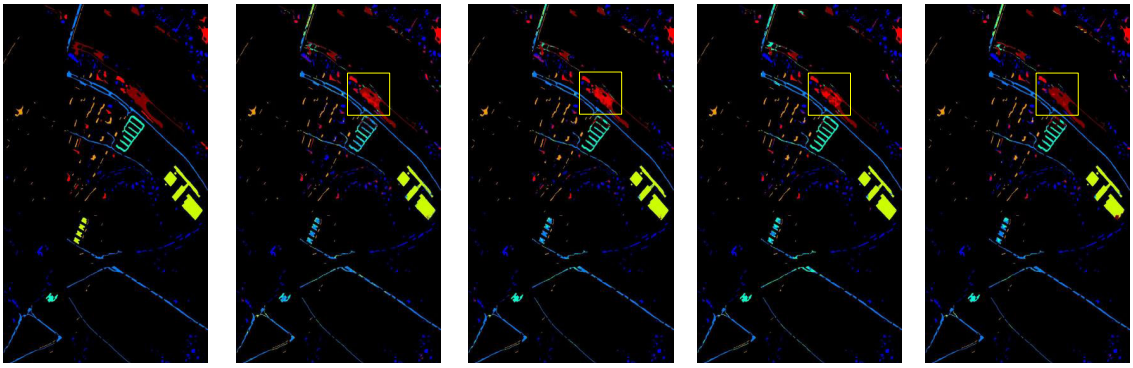


Fig. 11. Classification maps for target scene PC dataset obtained using two baselines and corresponding FSSG-improved methods. (a) Ground truth, (b) SDEnet (80.06%), (c) SDEnet + FSSG (83.63%), (d) S2AMSnet (83.36%), and (e) S2AMSnet + FSSG (86.27%).

2) *Analysis of Disentanglement Mechanism:* Column 5 of Tables V–VII provides cross-scene HSI classification results from DSDG, where a disentanglement mechanism was designed for extracting domain-invariant features. This method achieves competitive results compared to SDEnet and S^{2}MASnet. However, the S^{2}MASnet + FSSG outperforms it on three datasets, especially in the target scene Loukia (greater than five points gain over DSDG in terms of OA and KC). It indicates that feature disentanglement can effectively decompose the original data into domain-invariant features and domain-specific features. Moreover, applying it in the data expansion stage is more reasonable for single-source DG tasks, as ensuring effective and reasonable domain expansion is a prerequisite for improving generalization performance.

3) *Comparisons With General DG Methods:* Columns 2–3 of Tables V–VII list the classification scores of the general DG methods (PDEN and LDSG) in the three target scenes. The generalizability of these methods for HSI data is much lower than those that take the spectrum into account. This verifies that as the scene changes, even data from the same sensor suffer from domain shift problems. As a result, the design of FSSG in spectral is reasonable and necessary.

TABLE VIII

CLASS-SPECIFIC AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR THE TARGET SCENE HANCHUAN DATA

Class	Classification algorithms			
	SDEnet[38]	SDEnet+**FSSG**	S^{2}MAsnet[39]	S^{2}MASnet+**FSSG**
1	92.86	92.81	**95.02**	94.98
2	85.23	**91.41**	80.65	80.14
3	29.96	35.15	63.83	**68.6**
OA (%)	73.42±1.22	76.35±**0.78** (+2.93)	82.82±4.75	**84.0**±3.96 (+1.18)
KC (k)	58.61±2.05	63.23±**1.18** (+4.62)	72.9±7.45	**74.72**±2.41 (+1.35)

[1] The best and second-best results are highlighted in **bold** and gray

4) *Stability Analysis:* As shown by the standard deviation in each classification result, FSSG reduces the standard deviation of repeated independent experiments, which leads to a more stable generalization performance. Upon investigation, the pixelwise semantic consistency constraint in FSSG enhances robust context semantics and prevents overemphasis on domain-specific patterns during the training stage.

2) Qualitative Results: Figs. 10–13 show qualitative comparisons of SDEnet + **FSSG**/S^{2}MASnet + **FSSG** against SDEnet/S^{2}MASnet on four datasets (i.e., Houston, Pavia,

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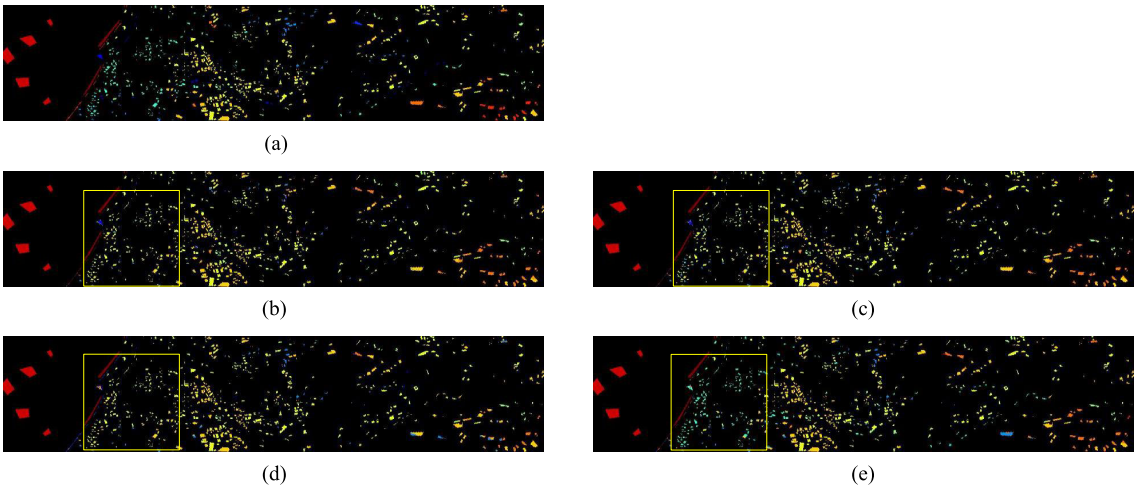


Fig. 12. Classification maps for target scene Loukia dataset obtained using two baselines and corresponding FSSG-improved methods. (a) Ground truth, (b) SDEnet (60.22%), (c) SDEnet + FSSG (62.45%), (d) S2AMSnet (64.51%), and (e) S2AMSnet + FSSG (67.53%).

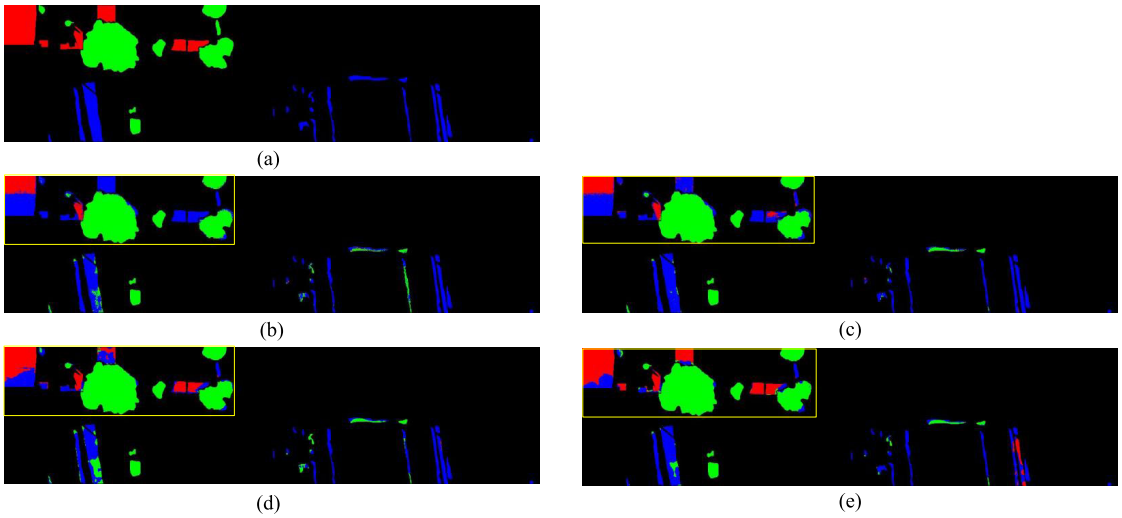


Fig. 13. Classification maps for target scene HanChuan dataset obtained using two baselines and corresponding FSSG-improved methods. (a) Ground truth, (b) SDEnet (73.42%), (c) SDEnet + FSSG (76.35%), (d) S2AMSnet (82.82%), and (e) S2AMSnet + FSSG (84.0%).

HyRANK, and WHU-Hi). As seen, FSSG demonstrates the ability to produce a more accurate prediction map in various cross-scene HSI classification tasks. For example, the first, third, and fifth categories (i.e., Grass healthy, Trees, and Residential buildings) in the Houston datasets; the first, fifth, and seventh categories (i.e., Trees, Bitumen, and Bare soil) in the Pavia datasets; the fourth and fifth categories (i.e., Fruit Trees and Olive Groves) in the HyRANK datasets; the second and third categories (i.e., Bare Soil and Red roof) in the WHU-Hi datasets.

C. Diagnostic Experiment

We further investigate the efficacy of our core ideas and the contribution of key components across the three introduced datasets. To perform extensive ablation experiments, each model was trained for 400 iterations and repeated ten times independently while keeping other hyperparameters unchanged.

1) Contribution of the Spectral and Spatial Disturbance: Ablation experiments are conducted by selecting only one of SpeE and SpaE to participate in the domain expansion module. Table IX presents the classification results of FSSG and its variant forms across the three datasets. Obviously, isolating SpeDB and SpaDB results in a decrease in the generalization performance of FSSG. The best performance occurs when both disturbances are conducted simultaneously, demonstrating the efficacy of the two disturbance mechanisms.

2) Semantic Expansion: To address the issue of invalid SD semantics in TD caused by changes in land-cover distribution, a semantic expansion module is designed to exchange semantics across HSI cubes of the same category, thereby generating more robust GD samples. An ablation study is conducted to validate this design, with results summarized in Fig. 14(a). Based on FSSG, a variant FSSG w/o phase disturbance (PDB) is derived: only focus on the changes in style without semantic in spatial. It obtains 81.01%, 84.42%, and 65.57% OA in three datasets respectively. Then, with the provision of PDB,

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TABLE IX
ABLATION STUDY OF SPEE AND SPAE MECHANISM IN FSSG

FSSG		Target Scene OA (%)		
SpaE	SpeE	Houston 2018	PC	Loukia
Baseline		SDEnet + FSSG		
w/o	w/	77.81	82.45	60.69
w/	w/o	79.26	80.73	61.64
w/	w/	**79.46**	**83.63**	**62.45**
		S^2MASnet + FSSG		
w/o	w/	80.33	85.49	61.77
w/	w/o	81.05	84.56	61.92
w/	w/	**82.06**	**86.27**	**67.53**

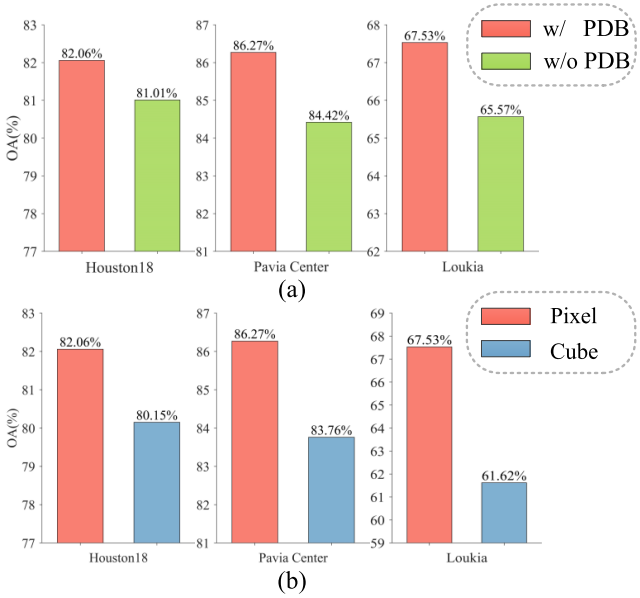


Fig. 14. Ablation study on components in SpaE and SpeE. (a) Generalization performance of FSSG with or without the PDB in SpaE. (b) Generalization performance of FSSG with pixelwise disturbance or cubewise disturbance in SpeE.

consistent performance gains (81.01% → 82.06% for Houston 2018, 84.42% → 86.27%, and 65.57% → 67.53%) can be observed. This evidence that exchanging semantics across HSI cubes of the same category results in a more effective extended distribution.

3) Pixelwise Versus Cubewise: To validate the design of the proposed pixelwise disturbance mechanism in spectral, the perturbed object (i.e., spectral pixel) is replaced with the HSI cube to generate a variant for comparison. As shown in Fig. 14(b), relaxing the semantic consistency constraint from the HSI cube to each pixel in spectral leads to a substantial performance gain.

4) Importance of Continuous Interpolation in Frequency Space: To analyze the benefits of continuous interpolation, the effects of α and β on model performance under different settings are discussed. Ablation experiments are conducted with fixed values from 0.0 to 1.0 in increments of 0.2, and continuous sampling within the range [0.0, 1.0]. Fig. 15 shows performance comparison of continued sampling against 6 fixed values on the PC. When $\alpha, \beta = 0$, a special case arises where FSSG operates without spatial disturbance. Compared

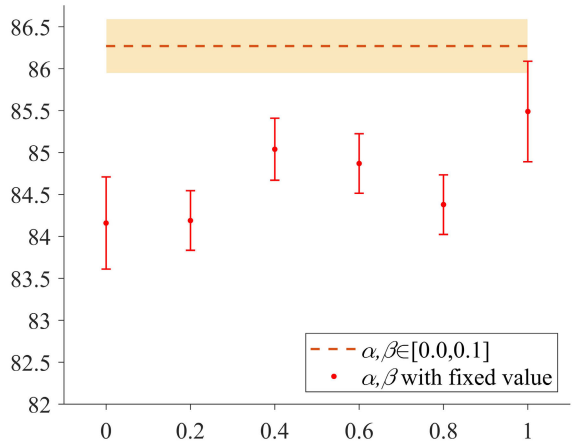


Fig. 15. Generalization performance on Pavia scene under different settings of interpolation ratios α, β, with fixed value or continuous sampling from different ranges (with error bar from ten independent runs).

TABLE X
NUMBER OF TRAINABLE PARAMETERS AND AVERAGE TRAINING TIME (IN SECONDS) FOR EACH EPOCH ON THREE DATASETS

Methods	SDEnet	SDEnet+**FSSG**	S^2MAsnet[39]	S^2MASnet+**FSSG**
Houston 2018				
Params(M)	1.88	1.61	1.07	1.04
Time(s)	4.57	6.10	7.84	10.43
Pavia Center				
Params(M)	2.53	2.12	0.94	0.89
Time(s)	18.82	20.34	20.08	25.14
Loukia				
Params(M)	3.42	2.82	0.97	0.91
Time(s)	10.89	13.91	11.05	13.16

to this case, any setting of $\alpha, \beta > 0$ consistently improves performance. Furthermore, continuous sampling in the range of [0.0, 1.0] yields additional gains, which evidences that continuous interpolation in frequency space facilitates more effective domain extension.

D. Time Consumption and Model Complexity

The integration of the FSSG demonstrates significant advantages in parameter efficiency and generalization capability for cross-scene HSI classification. As shown in Table X, the parameter size (Params) of the SDEnet series decreases consistently across three datasets after incorporating FSSG (e.g., from 1.88 M to 1.61 M in Houston2018), which can be attributed to the Fourier disentanglement mechanism. This mechanism decouples style and semantic components through amplitude–phase separation, where amplitude spectrum sharing preserves class-specific semantics while eliminating redundant parameters in traditional cubewise randomization. For the S2AMSnet series, the parameter reduction (e.g., 1.07 → 1.04 M) further validates that the continuous frequency–space interpolation dynamically adjusts spectral–spatial distributions through learnable mixing coefficients, reducing dependence on fixed convolution kernels.

Notably, the computational cost exhibits an acceptable trade-off—though training time increases by 33.5% (4.57 → 6.10 s for SDENet + FSSG) due to Fourier transforms

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and spectral–spatial perturbations, the classification accuracy improves 4.2%–6.8% across datasets. This verifies the effectiveness of FSSG's coarse-to-fine adversarial framework with relaxation procedure, which enhances context semantic robustness through pixelwise consistency constraints.

VI. Conclusion

The FSSG is introduced as a novel framework to generate diverse GDs to simulate the interdomain gap between an unseen domain and SD. Specifically, a Fourier-based disentanglement mechanism is first designed to decouple the original features into semantic and stylistic features from the spectral and spatial dimensions, respectively. By sharing semantic information in the frequency space of each HSI pixel, it ensures the generation of semantically consistent samples. In the spatial dimension, an effective continuous frequency–space interpolation mechanism is employed to transmit styles and context semantics across cubes, thereby creating a continuous distribution space to enhance generalizability. The superior efficacy of the proposed FSSG is validated through three public cross-scene HSI classification tasks. While the current framework addresses spectral and spatial dimensions separately, future work will explore unified Fourier-based spectral–spatial disentanglement mechanisms for cross-dimensional generalization.

Appendix A
Theoretical Proof of Phase Perturbation Stability

Let the spectral signal of an HSI pixel be $x \in \mathbb{R}^{1\times1\times C}$, its frequency space signal is obtained via a Fourier transform as follows: $\mathcal{F}(x)(u) = \sum_{c=0}^{C-1} x(c)e^{-j2\pi((c/C)u)}$. The amplitude $a = |\mathcal{F}(x)(u)|$ and phase $p = \angle\mathcal{F}(x)$ are then used to represent style and semantic information, respectively.

In SpeE, the concept of differential privacy is introduced, and noise conforming to a Laplace distribution is added to each sample in the style information bank to generate a new phase spectrum, which is represented as follows:

$$\widehat{p} = p + L(\varepsilon) \odot m \tag{17}$$

where $L(\varepsilon) \sim \text{Lap}(\varepsilon) = 1/(2\varepsilon)\exp(-(|L|)/\varepsilon)$ represents privacy-preserving noise, $m = \mathbb{1}_{c\in[-\lambda C:\lambda C]}$ is a binary mask that controls the scale of low-frequency component within the phase spectrum to be perturbed, and $\odot$ denotes Hadamard product.

Once the perturbed phase spectrum $\widehat{p}$ is obtained, it is combined with the original amplitude spectrum to generate a spectral pixel of GD through a 1-D inverse Fourier transform, as follows:

$$\widehat{x} = \mathcal{F}_1^{-1}\left(a \odot e^{j\widehat{p}}\right). \tag{18}$$

Since the amplitude spectrum remains intact and the phase perturbation is bounded (i.e., $|L(\varepsilon)| \ll \pi \Rightarrow \|e^{j(q+\mathbf{L})}\|_2 \leq (2(1-\cos\varepsilon))^{1/2}$), we can draw the following theorem:

Theorem 1 (Reconstruction Error Bound): For $\varepsilon \leq 0.1\pi$, relative to the total signal energy E_x, we have

$$E_p/E_x = \frac{\left\|\hat{x}-x\right\|_2}{\|x\|_2} \leq \frac{\varepsilon\sqrt{C}}{\sum_u a(u)} < 0.01 \quad (\text{when } \varepsilon = 0.05\pi) \tag{19}$$

where E_p denotes the perturbation energy.

Proof: When ε is appropriately set (typically, $\varepsilon \leq 0.1\pi$), the following approximation holds:

$$\mathcal{F}^{-1}\left(ae^{j(p+L)}\right) \approx \mathcal{F}^{-1}\left(ae^{jp}\right) + \mathcal{F}^{-1}\left(jaLe^{jp}\right). \tag{20}$$

The perturbation energy satisfies $E_P = \|\mathcal{F}^{-1}(jaLe^{jp})\|_2 \leq \varepsilon\sqrt{C}$.

Appendix B
Privacy Protection via Lipschitz Constraint

Define Lipschitz constant $\mathcal{L}$ for perturbation operator. For $\forall x$, $\mathcal{L}$ satisfies

$$\|G(x+\delta) - G(x)\|_2 \leq \mathcal{L}\|\delta\|_2. \tag{21}$$

The perturbation process satisfies

$$\mathcal{L} = \sup\left\|\frac{\partial\hat{x}}{\partial L}\right\|_2 \leq (1/C)\sum|m(u)| \leq (2\lambda C)/C = 2\lambda. \tag{22}$$

With $\gamma = 0.2$, we achieve $\mathcal{L} = 0.4$, guaranteeing linear control over output variations.

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